

Hadoop Installation on Windows OS – Requirements and Configuration Guide

Prerequisites

1. Windows 10/11 (64-bit)
2. Java Development Kit (JDK 8 or later)
3. Hadoop binary package (e.g., Hadoop 3.x)
4. Winutils.exe matching the Hadoop version
5. Minimum 8 GB RAM (recommended)
6. At least 20 GB free disk space

Step 1: Install Java

Download and install JDK. Set JAVA_HOME environment variable and add %JAVA_HOME%\bin to the PATH variable.

Step 2: Download Hadoop

Download Hadoop, extract it to C:\hadoop, and verify the folder structure.

Step 3: Configure Environment Variables

Create HADOOP_HOME=C:\hadoop and add: %HADOOP_HOME%\bin and %HADOOP_HOME%\sbin to PATH.

Step 4: Add Winutils

Copy winutils.exe into C:\hadoop\bin.

Step 5: Configure Hadoop Files

Edit core-site.xml, hdfs-site.xml, mapred-site.xml, and yarn-site.xml. Configure namenode, datanode, and temporary directories.

Step 6: Configure JAVA_HOME in Hadoop

Open hadoop-env.cmd and set JAVA_HOME to your JDK installation path.

Step 7: Format NameNode

Run: hdfs namenode -format

Step 8: Start Hadoop Services

Start HDFS and YARN using start-dfs.cmd and start-yarn.cmd.

Verification

Open browser: NameNode UI: <http://localhost:9870>
ResourceManager UI: <http://localhost:8088>

Recommended Configuration

- RAM: 8–16 GB
- CPU: Dual-core or higher
- Storage: SSD preferred
- Java: JDK 8/11
- Hadoop: Version 3.x